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Database Systems

Design, Implementation, & Management



National Economics University

Faculty of Mathematical Economics

Database System Design, Implementation And Management Course – Database System

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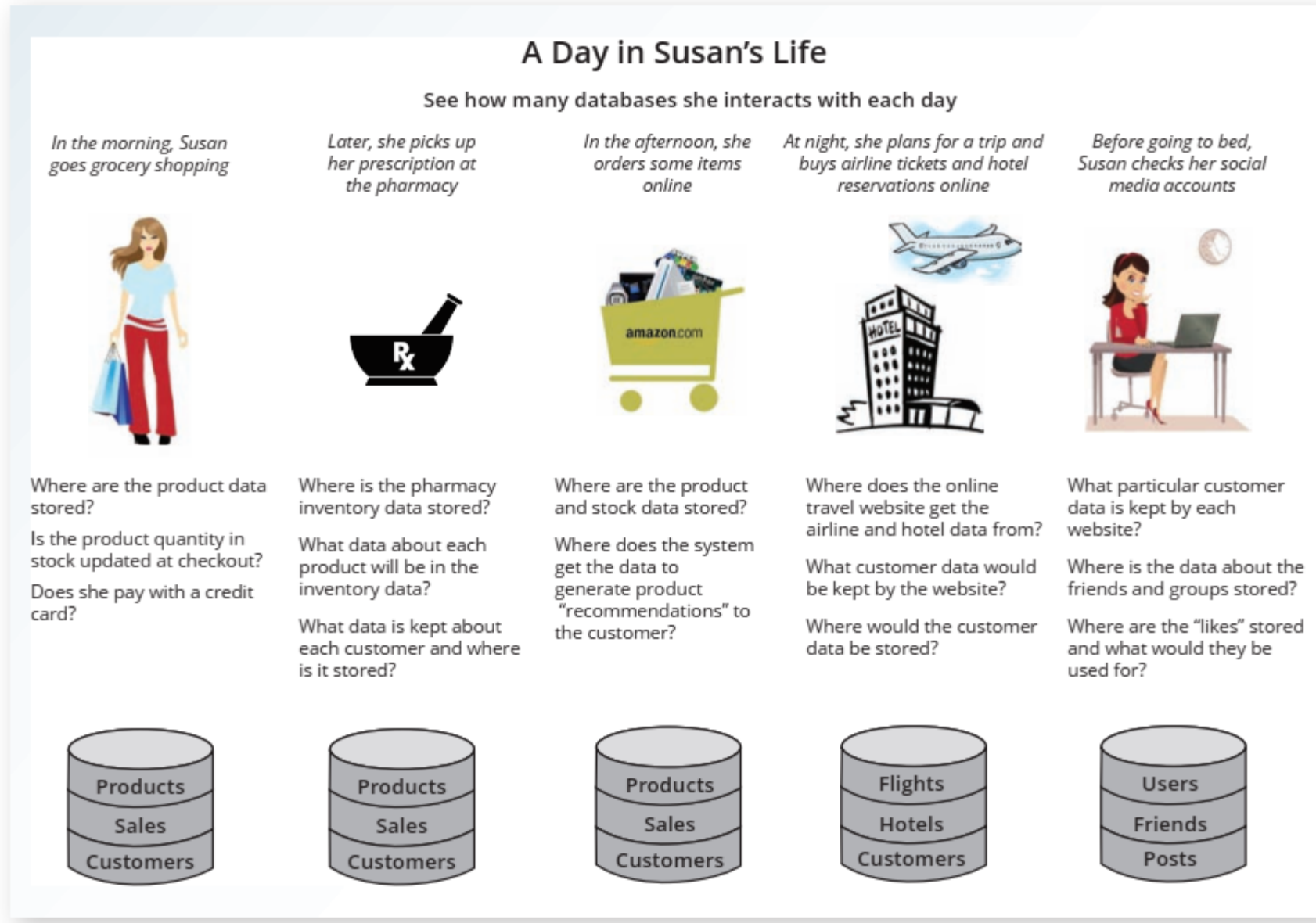
Learning Objectives

1. Define the difference between data and information
2. Describe what a database is, the various types of databases, and why they are valuable assets for decision making
3. Explain the importance of database design
4. Outline how modern databases evolved from file systems
5. Identify flaws in file system data management
6. Outline the main components of the database system
7. Describe the main functions of a database management system (DBMS)

Introduction

- ❑ Good decisions require good information derived from raw facts
- ❑ Data is managed most efficiently when stored in a database
- ❑ Databases evolved from computer file systems
- ❑ Understanding file system characteristics is important

Figure 1.1 The Pervasive Nature of Databases



1-1 Why Databases?

- ❑ Data is ubiquitous (abundant, global, everywhere) and pervasive (unescapable, prevalent, persistent).
 - From birth to death, we generate and consume data.
- ❑ Data is also essential for organizations to survive and prosper.
- ❑ Business information systems is to help businesses use information as an organizational resource.
 - At the heart of all of these systems are the collection, storage, aggregation, manipulation, dissemination, and management of data.
- ❑ Databases are specialized structures that allow computer-based systems to store, manage, and retrieve data very quickly.

1-2 Data versus Information

- ❑ Data are raw facts
- ❑ Information is the result of processing raw data to reveal meaning
- ❑ Information requires context to reveal meaning
- ❑ Raw data must be formatted for storage, processing, and presentation
- ❑ Data are the foundation of information, which is the bedrock of knowledge

1-2 Data versus Information (cont'd.)

- ❑ Data: building blocks of information
- ❑ Information produced by processing data
- ❑ Information used to reveal meaning in data
- ❑ Accurate, relevant, timely information is the key to good decision making
- ❑ Good decision making is the key to organizational survival
- ❑ Data management: focuses on proper generation, storage, and retrieval of data

data

Raw facts, or facts that have not yet been processed to reveal their meaning to the end user.

information

The result of processing raw data to reveal its meaning. Information consists of transformed data and facilitates decision making.

knowledge

The body of information and facts about a specific subject. Knowledge implies familiarity, awareness, and understanding of information as it applies to an environment. A key characteristic is that new knowledge can be derived from old knowledge.

Figure 1.2 Transforming Raw Data into Information

a) Data entry screen

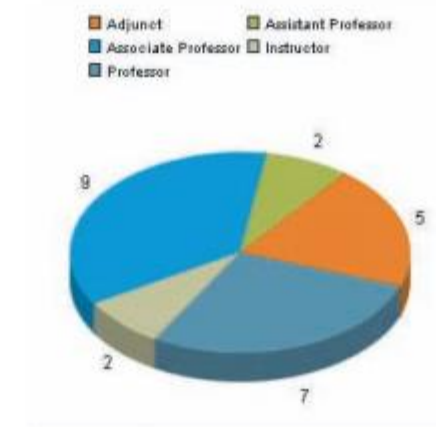
b) Raw data

ID	LastName	FirstName	DeptCode	Office	FEmail	Rank	HireYear	Degree
1	Washington	George	MGMT	N125	g.washington@mtsu.edu	Professor	2001	Ph.D.
2	Adams	John	FIN	N113	j.adams@mtsu.edu	Professor	1984	Ph.D.
3	Jefferson	L. Thomas	ECON		l.jefferson@mtsu.edu	Instructor	2002	M.S.A.
4	Madison	D. James	FIN	N126	j.madison@mtsu.edu	Associate Professor	1994	Ph.D.
5	Morse	N. James	ACCT	N401	j.morse@mtsu.edu	Assistant Professor	1995	Ph.D.
6	Adams	O. John	ACCT	N418	j.adams@mtsu.edu	Associate Professor	1989	Ph.D.
7	Jackson	C. Andrew	ECON	N383	a.jackson@mtsu.edu	Associate Professor	1999	Ph.D.
8	Van Buren	T. Mark	FIN	N106	m.vanburen@mtsu.edu	Professor	1980	Ph.D.
9	Harrison	R. William	MTGT	N118	w.harrison@mtsu.edu	Professor	1994	Ph.D.
10	Tyler	M. John	MGMT		j.tyler@mtsu.edu	Assistant Professor	2000	Ed.D.
11	Pull	Cheryl	MTGT	N143	c.pull@mtsu.edu	Associate Professor	2002	Ph.D.
12	Taylor	G. Zachary	ACCT	N415	z.taylor@mtsu.edu	Associate Professor	1996	Ph.D.
13	Pillmore	Michael	JCB	N219	m.pillmore@mtsu.edu	Professor	1992	Ph.D.
14	Pence	A. Franklin	MTGT	N159	f.pence@mtsu.edu	Instructor	2005	M.S.A.
15	Buchanan	T. James	MGMT	N146	j.buchanan@mtsu.edu	Associate Professor	1996	D.B.A.
17	Lynch	W. Larry	MGMT	N150	l.lynch@mtsu.edu	Associate Professor	1996	Ph.D.
18	Johnson	Andrew	SVS	N100	a.johnson@mtsu.edu	Professor	1987	Ph.D.
19	Gent	Kate	MTGT	N120	k.gent@mtsu.edu	Assistant Professor	1989	D.B.A.
20	Rutherford	Hayes	ACCT	N468	h.rutherford@mtsu.edu	Professor	1992	Ph.D.
21	Goffield	T. Denise	ACCT		d.goffield@mtsu.edu	Assistant Professor	2016	Ph.D.
22	Arthur	Emily	ACCT	N403	e.arthur@mtsu.edu	Associate Professor	2002	J.D.
23	Cleveland	G. Robert	ACCT	N401	r.cleveland@mtsu.edu	Associate Professor	1987	Ph.D.
24	Harrison	K. Patrick	BUSA	N406	p.harrison@mtsu.edu	Associate Professor	2001	J.D.
25	McKinley	B. Prescille	SVS	N383	b.mckinley@mtsu.edu	Adjunct	1994	M.S.
26	Russcove	F. Hilary	MGMT	N104	h.russcove@mtsu.edu	Associate Professor	2002	Ph.D.
27	Wilson	Lewis	SCEN	N440	l.wilson@mtsu.edu	Professor	1992	Ph.D.
28	Harding	William	MTGT	N114	w.harding@mtsu.edu	Professor	1994	Ed.D.
29	Cookidge	Calvin	ECON	N116	c.cookidge@mtsu.edu	Professor	1975	Ph.D.
30	Hoover	Lisa	MGMT		l.hoover@mtsu.edu	Adjunct	1978	M.S.A.
31	Truman	Betty	ACCT	N416	b.truman@mtsu.edu	Professor	1971	Ed.D.
32	Johnson	Robert	SCEN	N240	r.johnson@mtsu.edu	Professor	2001	Ph.D.

c) Information in summary format

Rank	COUNT	%INFS	TOT/COL	%/COL. TOT.	%/COL. FAC.
Adjunct	6	20.00%	23	21.74%	3.27%
Assistant Professor	2	8.00%	28	7.14%	1.31%
Associate Professor	9	36.00%	37	24.32%	5.88%
Instructor	2	8.00%	18	11.11%	1.31%
Professor	7	28.00%	47	14.89%	4.58%

d) Information in graphical format



1-3 Introducing the Database

❑ Database: shared, integrated computer structure that stores a collection of:

- End-user data: raw facts of interest to end user
- Metadata: data about data
 - ✓ Provides description of data characteristics and relationships in data
 - ✓ Complements and expands value of data

❑ Database management system (DBMS): collection of programs that manages the database structure and controls access to the data stored in the database

metadata

Data about data; that is, data about data characteristics and relationships. See also *data dictionary*.

database

A shared, integrated computer structure that houses a collection of related data. A database contains two types of data: end-user data (raw facts) and metadata.

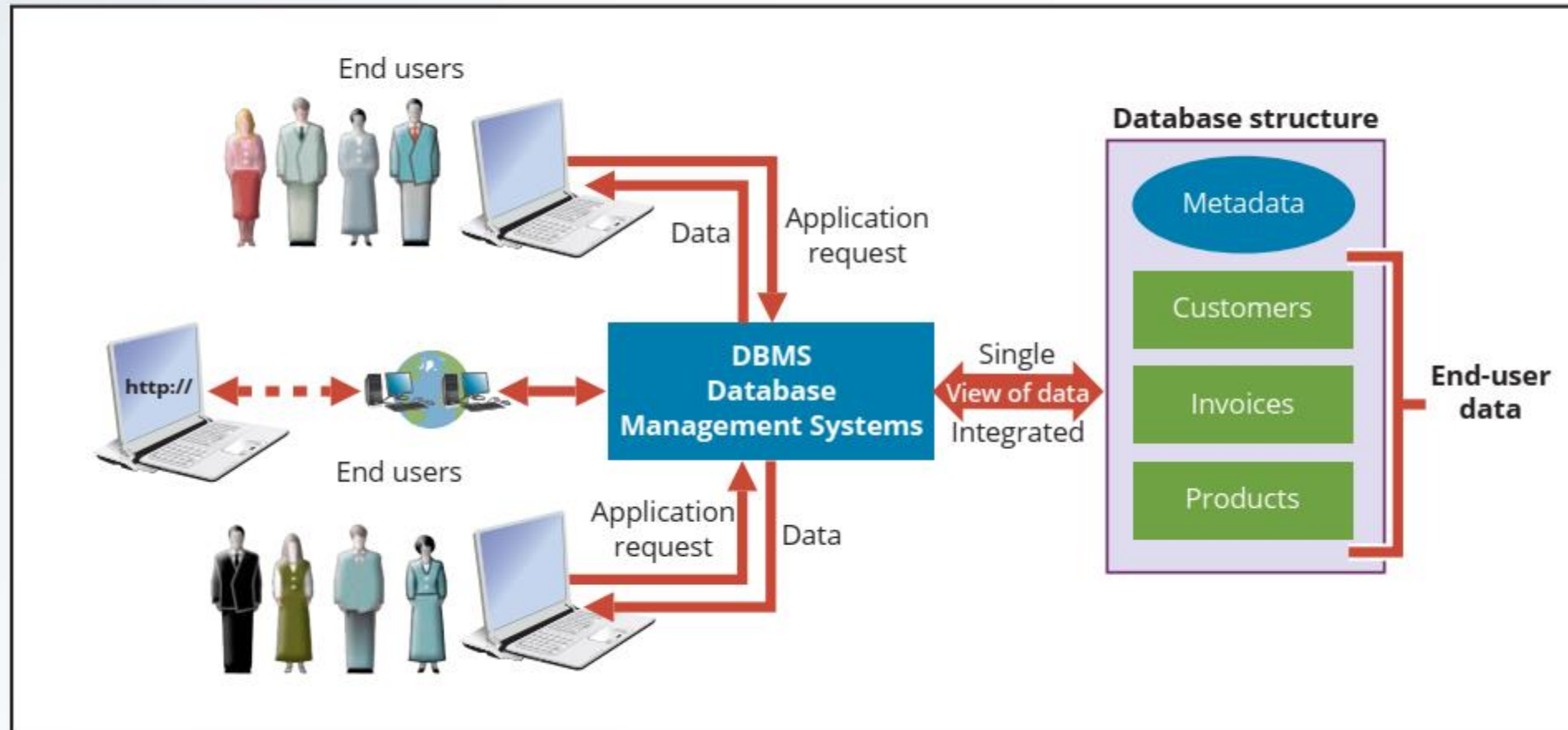
database management system (DBMS)

The collection of programs that manages the database structure and controls access to the data stored in the database.

1-3a Role and Advantages of the DBMS

- ❑ DBMS is the intermediary between the user and the database
 - Database structure stored as file collection
 - Can only access files through the DBMS
- ❑ DBMS enables data to be shared
- ❑ DBMS integrates many users' views of the data

Figure 1.4 The DBMS Manages the Interaction between the End User and the Database



Role and Advantages of the DBMS (cont'd.)

□ Advantages of a DBMS:

- Improved data sharing
- Improved data security
- Better data integration
- Minimized data inconsistency
- Improved data access
- Improved decision making
- Increased end-user productivity

data inconsistency

A condition in which different versions of the same data yield different (inconsistent) results.

query

A question or task asked by an end user of a database in the form of SQL code.
A specific request for data manipulation issued by the end user or the application to the DBMS.

data quality

A comprehensive approach to ensuring the accuracy, validity, and timeliness of data.

Types of Databases

- ❑ Databases can be classified according to:
 - Number of users
 - Database location(s)
 - Expected type and extent of use
- ❑ Single-user database supports only one user at a time
 - Desktop database: single-user; runs on PC
- ❑ Multiuser database supports multiple users at the same time
 - Workgroup and enterprise databases

single-user database

A database that supports only one user at a time.

desktop database

A single-user database that runs on a personal computer.

multiuser database

A database that supports multiple concurrent users.

workgroup database

A multiuser database that usually supports fewer than 50 users or is used for a specific department in an organization.

enterprise database

The overall company data representation, which provides support for present and expected future needs.

Types of Databases (cont'd.)

- ❑ Centralized database: data located at a single site
- ❑ Distributed database: data distributed across several different sites
- ❑ Cloud database: is a database that is created and maintained using cloud data services, such as Microsoft Azure or Amazon AWS

centralized database

A database located at a single site.

distributed database

A logically related database that is stored in two or more physically independent sites.

cloud database

A database that is created and maintained using cloud services, such as Microsoft Azure or Amazon AWS.

Types of Databases (cont'd.)

- ❑ General-purpose database: contain a wide variety of data used in multiple disciplines—for example, a census database that contains general demographic data and the LexisNexis and ProQuest databases that contain newspaper, magazine, and journal articles for a variety of topics.
- ❑ Discipline-specific databases: contain data focused on specific subject areas

general-purpose database

A database that contains a wide variety of data used in multiple disciplines.

discipline-specific database

A database that contains data focused on a specific subject area.

Types of Databases (cont'd.)

- ❑ Operational database - online transaction processing (OLTP) database – transactional database - or production database: a database that is designed primarily to support a company's day-to-day operations.
- ❑ Analytical database focuses primarily on storing historical data and business metrics used exclusively for tactical or strategic decision making.

operational database

A database designed primarily to support a company's day-to-day operations. Also known as a *transactional database*, *OLTP database*, or *production database*.

online transaction processing (OLTP) database

See operational database.

transactional database

See operational database.

production database

See operational database.

analytical database

A database focused primarily on storing historical data and business metrics used for tactical or strategic decision making.

Types of Databases (cont'd.)

- ❑ Unstructured data exist in their original state
- ❑ Structured data result from formatting
 - Structure applied based on type of processing to be performed
- ❑ Semi-structured data have been processed to some extent
- ❑ Extensible Markup Language (XML) represents data elements in textual format
 - XML database supports semi-structured XML data

unstructured data

Data that exists in its original, raw state; that is, in the format in which it was collected.

structured data

Data that has been formatted to facilitate storage, use, and information generation.

semistructured data

Data that has already been processed to some extent.

Extensible Markup Language (XML)

A metalanguage used to represent and manipulate data elements. Unlike other markup languages, XML permits the manipulation of a document's data elements.

XML database

A database system that stores and manages semistructured XML data.

social media

Web and mobile technologies that enable “anywhere, anytime, always on” human interactions.

NoSQL

A new generation of DBMS that is not based on the traditional relational database model.

Table 1.1 Types of Databases

Product	Number of Users			Data Location		Data Usage		XML
	Single User	Multiuser		Centralized	Distributed	Operational	Analytical	
		Workgroup	Enterprise					
MS Access	X	X		X		X		
MS SQL Server	X*	X	X	X	X	X	X	X
IBM DB2	X*	X	X	X	X	X	X	X
MySQL	X	X	X	X	X	X	X	X
Oracle RDBMS	X*	X	X	X	X	X	X	X

*Vendor offers single-user/personal or Express DBMS versions

1-4 Why Database Design Is Important

- ❑ Database design focuses on design of database structure used for end-user data
 - Designer must identify database's expected use
- ❑ Well-designed database:
 - Facilitates data management
 - Generates accurate and valuable information
- ❑ Poorly designed database:
 - Causes difficult-to-trace errors

database design

The process that yields the description of the database structure and determines the database components.

The second phase of the database life cycle.

Figure 1.5 Employee Skills Certification in a Poor Design

Why are there blanks in rows 9 and 10?

How to produce an alphabetical listing of employees?

How to count how many employees are certified in Basic Database Manipulation?

Is Basic Database Manipulation the same as Basic DB Manipulation?

What if an employee acquires a fourth certification?

Do we add another column?

ID	ENum	Name	Title	HireDate	Skill1	Skill1Date	Skill2	Skill2Date	Skill3	Skill3Date
1	02345	Brian Oates	DBA	2/14/2001	Basic Database Management	2/14/2008	Advanced Database Management	2/14/2011	Basic Web Design	8/9/2009
2	08273	Marco Bienz	Analyst	7/28/2012	Basic Web Design	3/8/2015	Advance Process Modeling	8/19/2018		
3	06234	Jasmine Patel	Programmer	8/10/2011	Basic Web Design	8/10/2013	Advanced C# programming	8/10/2013	Basic DB manipulation	1/29/2018
4	03373	Franklin Johnson, Jr.	Purchasing Agent	3/15/2008	Advanced Spreadsheets	6/20/2017				
5	13567	Almond, Robert	Analyst	9/30/2018	Basic Process Modeling	9/30/2020	Basic Database Design	5/23/2021		
6	10282	Richardson, Amanda	Clerk	4/11/2017						
7	09382	Susan Mathis	Database Programmer	8/2/2016	Basic DB Design	8/2/2018	Basic Database Manipulation	8/2/2018	Advanced DB Manipulation	5/1/2019
8	14311	Duong, Lee	Programmer	9/1/2020	Basic Web Design	9/1/2022				
9					Master Database Programming					
10					Basic Spreadsheets					
11	09002	Wade Gaither	Clerk	5/20/2016	Advanced Spreadsheets	5/16/2019	Basic Web Design	5/16/2019		
12	13383	Raymond F. Matthews	Programmer	3/12/2018	Basic C# Programming	3/12/2020				
13	09283	Chavez, Juan	Clerk	7/4/2016						
14	04893	Patricia Richards	DBA	6/11/2010	Advanced Database Management	6/11/2012	Advanced Database Manipulation	9/20/2018		
15	13932	Lee, Megan	Programmer	9/29/2019						

Figure 1.6 Employee Skills Certification in a Good Design

Table name: EMPLOYEE

Employee_ID	Employee_FName	Employee_LName	Employee_HireDate	Employee_Title
02345	Brian	Oates	2/14/2003	DBA
03373	Franklin	Johnson	3/15/2010	Purchasing Agent
04893	Patricia	Richards	6/11/2012	DBA
06234	Jasmine	Patel	8/10/2013	Programmer
06273	Marco	Bienz	7/28/2014	Analyst
09002	Wade	Gaither	5/20/2018	Clerk
09263	Juan	Chavez	7/4/2018	Clerk
09382	Susan	Mathis	8/2/2018	Database Programmer
10282	Amanda	Richardson	4/11/2019	Clerk
13383	Raymond	Matthews	3/12/2020	Programmer
13567	Robert	Almond	9/30/2020	Analyst
13932	Megan	Lee	9/29/2021	Programmer
14311	Lee	Duong	9/1/2022	Programmer

Table name: SKILL

Skill_ID	Skill_Name	Skill_Description
100	Basic Database Management	Create and manage database user accounts.
110	Basic Web Design	Create and maintain HTML and CSS documents.
120	Advanced Spreadsheets	Use of advanced functions, user-defined functions, and macroing.
130	Basic Process Modeling	Create core business process models using standard libraries.
140	Basic Database Design	Create simple data models.
150	Master Database Programming	Create integrated trigger and procedure packages for a distributed environ.....
160	Basic Spreadsheets	Create single tab worksheets with basic formulas
170	Basic C# Programming	Create single-tier data aware modules.
180	Advanced Database Management	Manage Database Server Clusters.
190	Advance Process Modeling	Evaluate and Redesign cross-functional internal and external business processes.
200	Advanced C# Programming	Create multi-tier applications using multi-threading
210	Basic Database Manipulation	Create simple data retrieval and manipulation statements in SQL.
220	Advanced Database Manipulation	Use of advanced data manipulation methods for multi-table inserts, set operations, and correlated subqueries.

Database name: Ch01_Text

Table name: CERTIFIED

Employee_ID	Skill_ID	Certified_Date
02345	100	2/14/2008
02345	110	8/9/2009
02345	180	2/14/2011
03373	120	6/20/2017
04893	180	6/11/2012
04893	220	9/20/2018
06234	110	8/10/2013
06234	200	8/10/2013
06234	210	1/29/2018

With the improved structure in Figure 1.6, you can use simple commands in a standard data manipulation language to do the following:

- Produce an alphabetical listing of employees by last name:

```
SELECT * FROM EMPLOYEE ORDER BY EMPLOYEE_LNAME;
```

- Determine how many employees are certified in Basic Database Manipulation:

```
SELECT COUNT(*)
```

```
FROM SKILL JOIN CERTIFIED ON SKILL.SKILL_ID = CERTIFIED.SKILL_ID
WHERE SKILL_NAME = 'Basic Database Manipulation';
```

1-5 Evolution of File System Data Processing

□ Reasons for studying file systems:

- Complexity of database design is easier to understand
- Understanding file system problems helps to avoid problems with DBMS systems
- Knowledge of file system is useful for converting file system to database system

□ File systems typically composed of collection of file folders, each tagged and kept in cabinet

- Organized by expected use

data processing (DP) specialist

The person responsible for developing and managing a computerized file processing system.

field

A character or group of characters (alphabetic or numeric) that has a specific meaning. A field is used to define and store data.

record

A logically connected set of one or more fields that describes a person, place, or thing.

file

A collection of related records. For example, a file might contain data about the students currently enrolled at Gigantic University.

1-5 Evolution of File System Data Processing (cont'd.)

- ❑ Contents of each file folder are logically related
- ❑ Manual file systems
 - Served as a data repository for small data collections
 - Cumbersome for large collections
- ❑ Computerized file systems
 - Data processing (DP) specialist converted computer file structure from manual system
 - ✓ Wrote software that managed the data
 - ✓ Designed the application programs

1-5 Evolution of File System Data Processing (cont'd.)

- ❑ Initially, computer file systems resembled manual systems
- ❑ As number of files increased, file systems evolved
 - Each file used its own application program to store, retrieve, and modify data
 - Each file was owned by individual or department that commissioned its creation

Figure 1.7 Contents of the Customer File

Database name: Ch01_Text

C_NAME	C_PHONE	C_ADDRESS	C_ZIP	A_NAME	A_PHONE	TP	AMT	REN
Alfred A. Ramas	615-844-2573	218 Fork Rd., Babs, TN	36123	Leah F. Hahn	615-882-1244	T1	100.00	05-Apr-2022
Leona K. Dunne	713-894-1238	Box 12A, Fox, KY	25246	Alex B. Alby	713-228-1249	T1	250.00	16-Jun-2022
Kathy W. Smith	615-894-2285	125 Oak Ln, Babs, TN	36123	Leah F. Hahn	615-882-2144	S2	150.00	29-Jan-2023
Paul F. Olowski	615-894-2180	217 Lee Ln., Babs, TN	36123	Leah F. Hahn	615-882-1244	S1	300.00	14-Oct-2022
Myron Orlando	615-222-1672	Box 111, New, TN	36155	Alex B. Alby	713-228-1249	T1	100.00	28-Dec-2022
Amy B. O'Brian	713-442-3381	387 Troll Dr., Fox, KY	25246	John T. Okon	615-123-5589	T2	850.00	22-Sep-2022
James G. Brown	615-297-1228	21 Tye Rd., Nash, TN	37118	Leah F. Hahn	615-882-1244	S1	120.00	25-Mar-2023
George Williams	615-290-2556	155 Maple, Nash, TN	37119	John T. Okon	615-123-5589	S1	250.00	17-Jul-2022
Anne G. Farriss	713-382-7185	2119 Elm, Crew, KY	25432	Alex B. Alby	713-228-1249	T2	100.00	03-Dec-2022
Olette K. Smith	615-297-3809	2782 Main, Nash, TN	37118	John T. Okon	615-123-5589	S2	500.00	14-Mar-2023

C_NAME = Customer name
C_PHONE = Customer phone
C_ADDRESS = Customer address
C_ZIP = Customer zip code

A_NAME = Agent name
A_PHONE = Agent phone
TP = Insurance type
AMT = Insurance policy amount, in thousands of \$
REN = Insurance renewal date

Table 1.2 Basic File Terminology

Term	Definition
Data	Raw facts, such as a telephone number, a birth date, a customer name, and a year-to-date (YTD) sales value. Data has little meaning unless it has been organized in some logical manner.
Field	A character or group of characters (alphabetic or numeric) that has a specific meaning. A field is used to define and store data.
Record	A logically connected set of one or more fields that describes a person, place, or thing. For example, the fields that constitute a record for a customer might consist of the customer's name, address, phone number, date of birth, credit limit, and unpaid balance.
File	A collection of related records. For example, a file might contain data about the students currently enrolled at Gigantic University.

Figure 1.8 Contents of the Agent File

A_NAME	A_PHONE	A_ADDRESS	ZIP	HIRED	YTD_PAY	YTD_FIT	YTD_FICA	YTD_SLS	DEP
Alex B. Alby	713-228-1249	123 Toll, Nash, TN	37119	01-Nov-2004	26566.24	6641.56	2125.30	132737.75	3
Leah F. Hahn	615-882-1244	334 Main, Fox, KY	25246	23-May-1990	32213.78	8053.44	2577.10	138967.35	0
John T. Okon	615-123-5589	452 Elm, New, TN	36155	15-Jun-2009	23198.29	5799.57	1855.86	127093.45	2

A_NAME = Agent name

A_PHONE = Agent phone

A_ADDRESS = Agent address

ZIP = Agent zip code

HIRED = Agent date of hire

YTD_PAY = Year-to-date pay

YTD_FIT = Year-to-date federal income tax paid

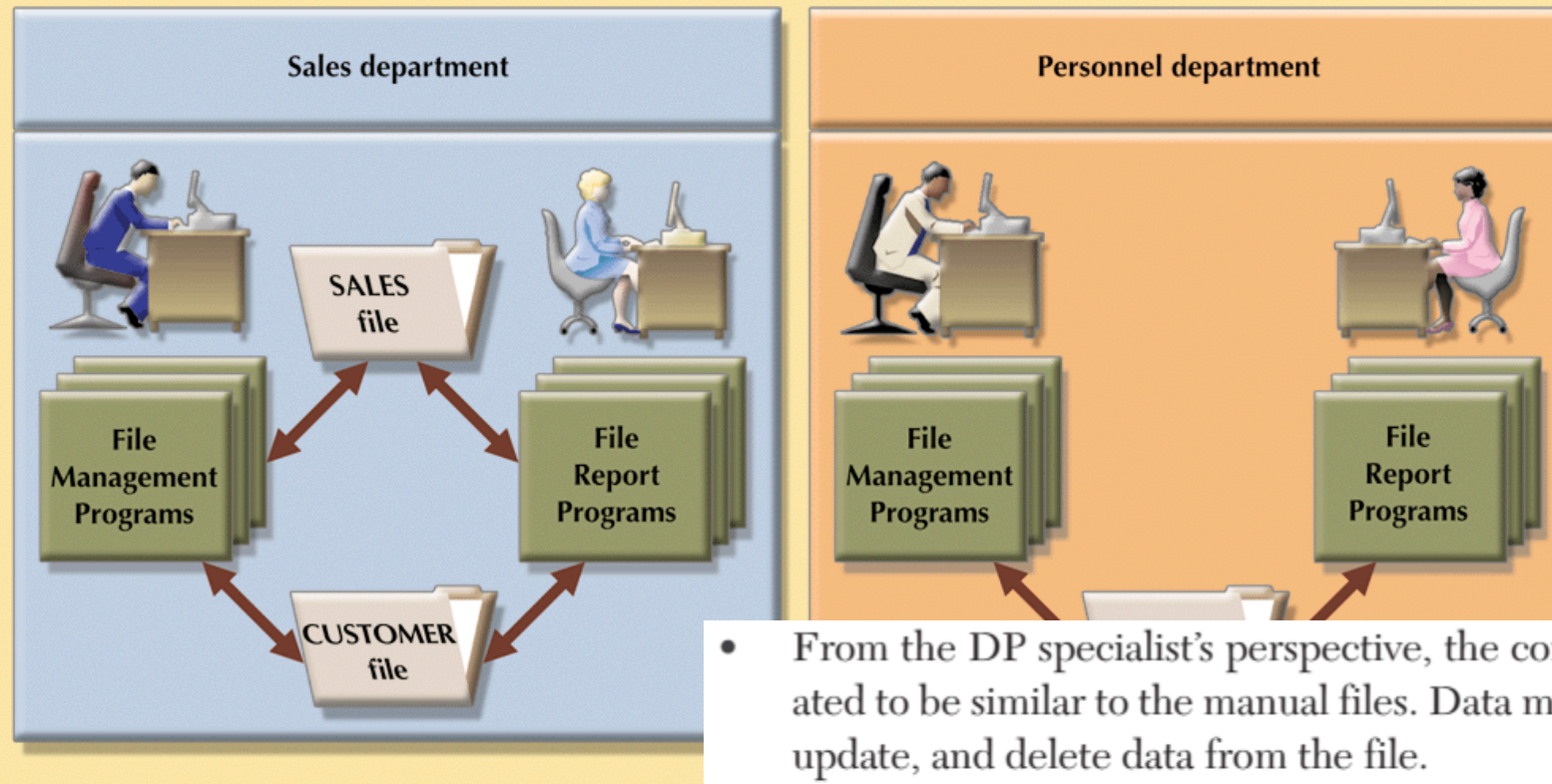
YTD_FICA = Year-to-date Social Security taxes paid

YTD_SLS = Year-to-date sales

DEP = Number of dependents

FIGURE
1.7

A simple file system



- From the DP specialist's perspective, the computer files within the file system were created to be similar to the manual files. Data management programs were created to add to, update, and delete data from the file.
- From the end user's perspective, the systems separated the users from the data. As the users' competitive environment pushed them to make more and more decisions in less time, users became frustrated by the delay between conceiving of a new way to create information from the data and the point when the DP specialist actually created the programs to generate that information.

File System Redux: Modern End-User Productivity Tools

- ❑ Ubiquitous use of personal productivity tools can introduce the same problems as the old file systems
- ❑ Microsoft Excel
 - Widely used by business users
 - Users have become so adept at working with spreadsheets, they tend to use them to complete tasks for which spreadsheets are not appropriate – database substitute

1-6 Problems with File System Data Processing

- ❑ File systems were an improvement over manual system
 - File systems used for more than two decades
 - Understanding the shortcomings of file systems aids in development of modern databases
 - Many problems not unique to file systems
- ❑ Even simple file system retrieval task required extensive programming
 - Ad hoc queries impossible
 - Changing existing structure difficult

1-6 Problems with File System Data Processing (cont'd.)

- ❑ Security features difficult to program
 - Often omitted in file system environments
- ❑ Summary of file system limitations:
 - Requires extensive programming
 - Cannot perform ad hoc queries
 - System administration is complex and difficult
 - Difficult to make changes to existing structures
 - Security features are likely to be inadequate

1-6a Structural and Data Dependence

- ❑ Structural dependence: access to a file is dependent on its own structure
 - All file system programs must be modified to conform to a new file structure
- ❑ Structural independence: change file structure without affecting data access
- ❑ Data dependence: data access changes when data storage characteristics change
- ❑ Data independence: data storage characteristics do not affect data access

structural dependence

A data characteristic in which a change in the database schema affects data access, thus requiring changes in all access programs.

structural independence

A data characteristic in which changes in the database schema do not affect data access.

data type

Defines the kind of values that can be used or stored. Also, used in programming languages and database systems to determine the operations that can be applied to such data.

data dependence

A data condition in which data representation and manipulation are dependent on the physical data storage characteristics.

data independence

A condition in which data access is unaffected by changes in the physical data storage characteristics.

logical data format

The way a person views data within the context of a problem domain.

physical data format

The way a computer “sees” (stores) data.

1-6a Structural and Data Dependence (cont'd.)

- ❑ Practical significance of data dependence is difference between logical and physical format
- ❑ Logical data format: how human views the data
- ❑ Physical data format: how computer must work with data
- ❑ Each program must contain:
 - Lines specifying opening of specific file type
 - Record specification
 - Field definitions

1-6b Data Redundancy

- ❑ File system structure makes it difficult to combine data from multiple sources
 - Vulnerable to security breaches
- ❑ Organizational structure promotes storage of same data in different locations
 - Islands of information
- ❑ Data stored in different locations is unlikely to be updated consistently
- ❑ Data redundancy: same data stored unnecessarily in different places

islands of information

In the old file system environment, pools of independent, often duplicated, and inconsistent data created and managed by different departments.

data redundancy

Exists when the same data is stored unnecessarily at different places.

1-6b Data Redundancy (cont'd.)

- ❑ Data inconsistency: different and conflicting versions of same data occur at different places
- ❑ Poor data security
- ❑ Data inconsistency
- ❑ Data-entry errors
- ❑ Data integrity problems

1-6c Data Anomalies

❑ Data anomalies: abnormalities when all changes in redundant data are not made correctly

- Update anomalies
- Insertion anomalies
- Deletion anomalies

data anomaly

A data abnormality in which inconsistent changes have been made to a database. For example, an employee moves, but the address change is not corrected in all files in the database.

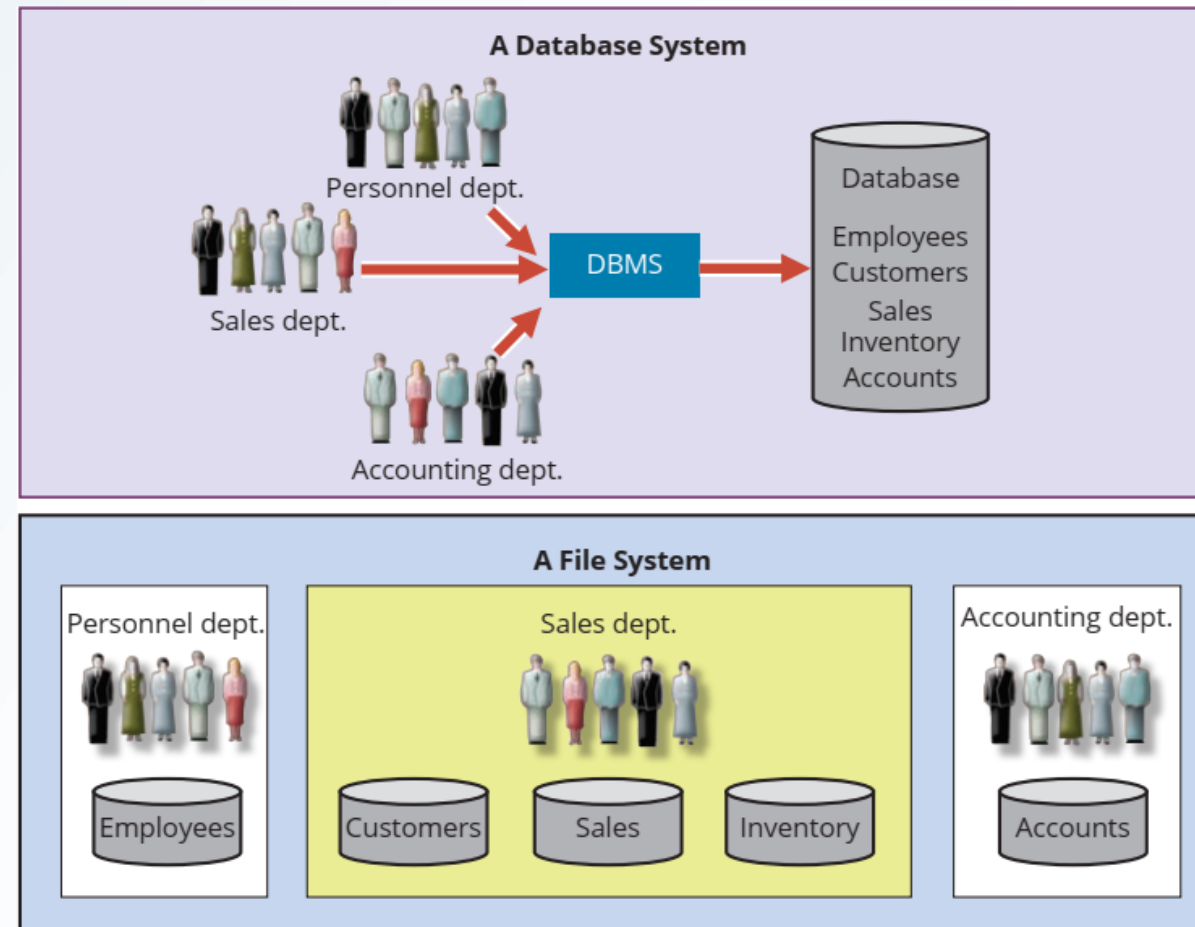
1-7 Database Systems

- ❑ Database system consists of logically related data stored in a single logical data repository
 - May be physically distributed among multiple storage facilities
 - DBMS eliminates most of file system's problems
 - Current generation stores data structures, relationships between structures, and access paths
 - ✓ Also defines, stores, and manages all access paths and components

database system

An organization of components that defines and regulates the collection, storage, management, and use of data in a database environment.

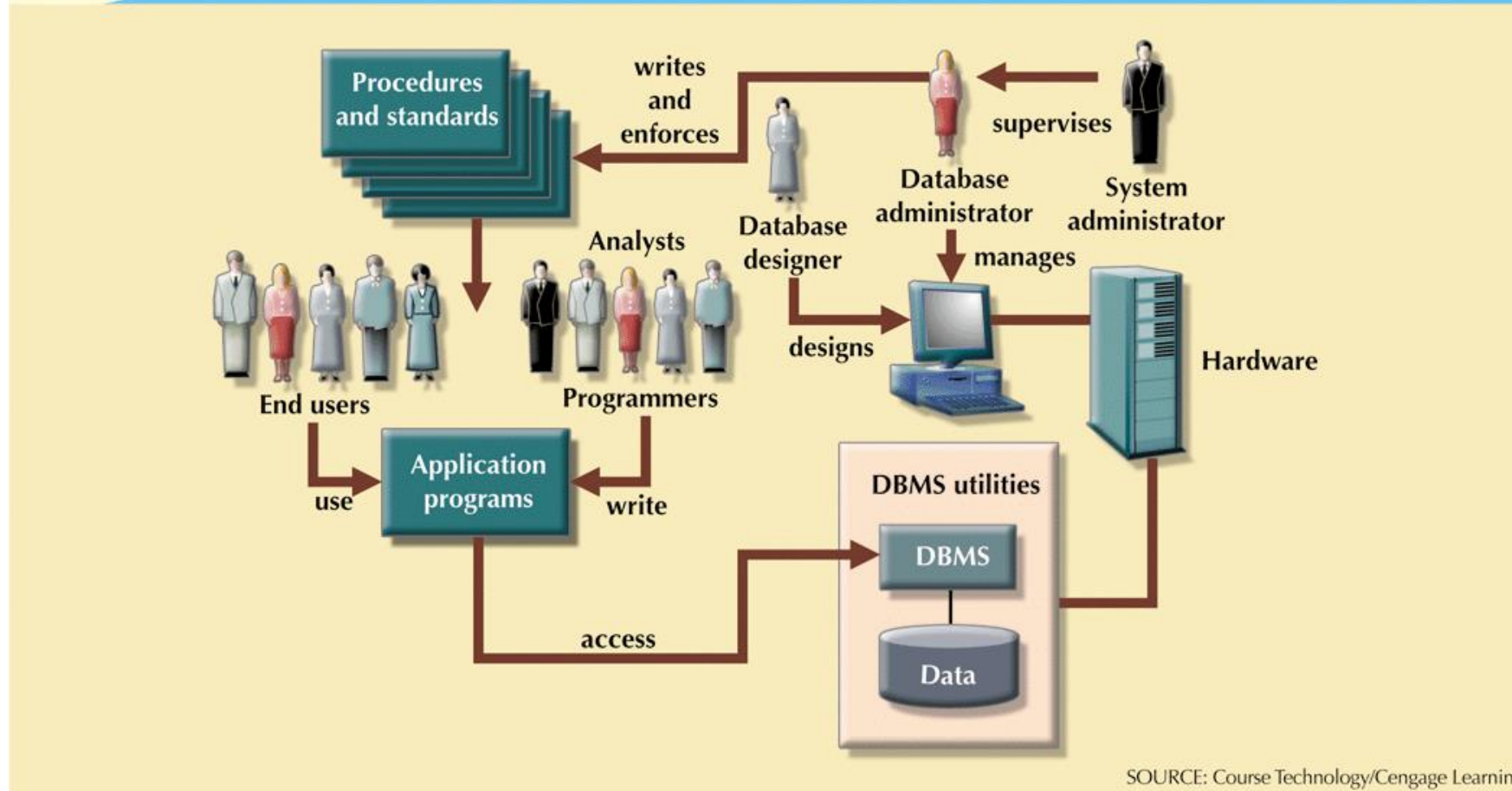
Figure 1.9 Contrasting Database and File Systems



1-7a The Database System Environment

- ❑ Database system: defines and regulates the collection, storage, management, use of data
- ❑ Five major parts of a database system:
 - Hardware
 - Software
 - People
 - Procedures
 - Data

FIGURE 1.9 The database system environment



1-7a The Database System Environment (cont'd.)

- ❑ Hardware: all the system's physical devices
- ❑ Software: three types of software required
 - Operating system software
 - DBMS software
 - Application programs and utility software

1-7a The Database System Environment (cont'd.)

- ❑ People: all users of the database system
 - System and database administrators
 - Database designers
 - Systems analysts and programmers
 - End users
- ❑ Procedures: instructions and rules that govern the design and use of the database system
- ❑ Data: the collection of facts stored in the database

1-7a The Database System Environment (cont'd.)

- ❑ Database systems are created and managed at different levels of complexity
- ❑ Database solutions must be cost-effective as well as tactically and strategically effective
- ❑ Database technology already in use affects selection of a database system

1-7b DBMS Functions

- ❑ Most functions are transparent to end users
 - Can only be achieved through the DBMS
- ❑ Data dictionary management
 - DBMS stores definitions of data elements and relationships (metadata) in a data dictionary
 - DBMS looks up required data component structures and relationships
 - Changes automatically recorded in the dictionary
 - DBMS provides data abstraction and removes structural and data dependency

data dictionary

A DBMS component that stores metadata—data about data. The data dictionary contains data definitions as well as data characteristics and relationships. May also include data that is external to the DBMS.

performance tuning

Activities that make a database perform more efficiently in terms of storage and access speed.

Figure 1.11 Illustrating Metadata with Microsoft SQL Server Express

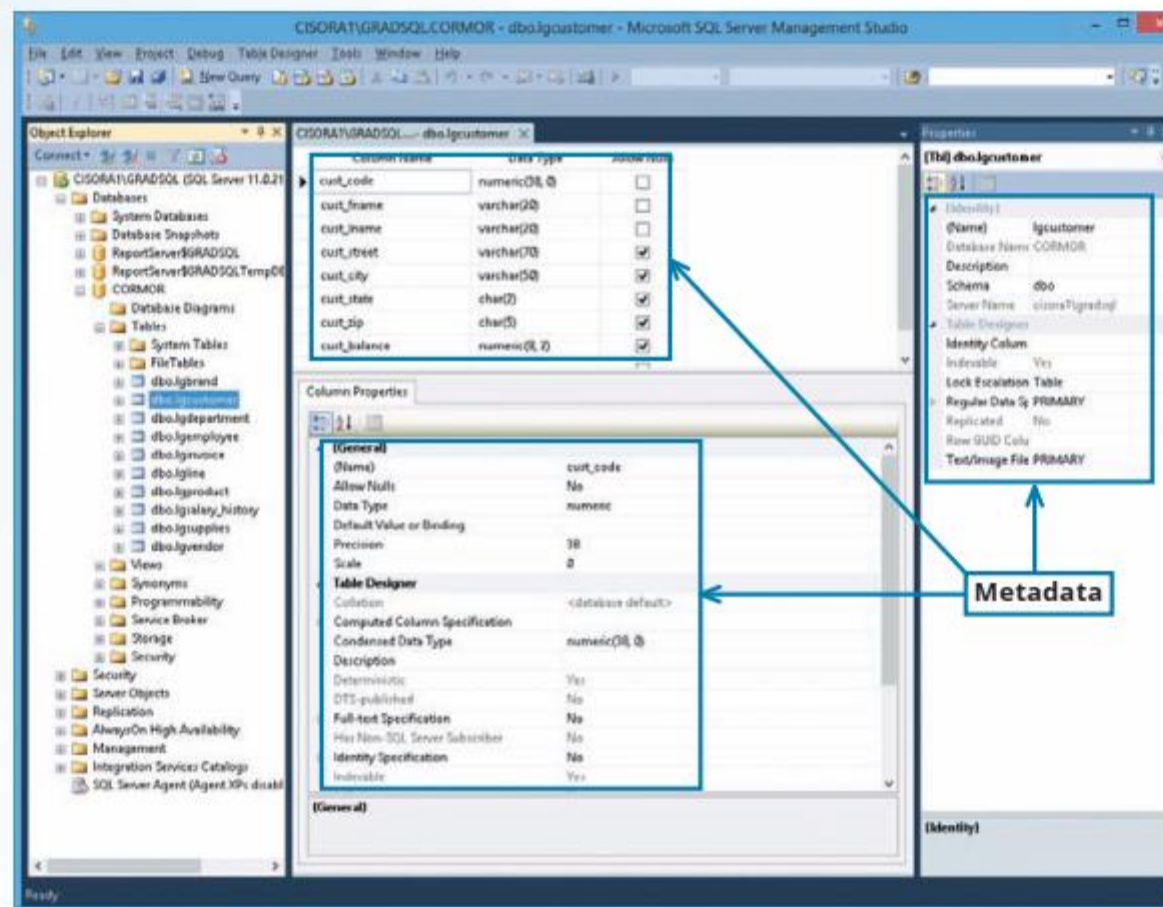
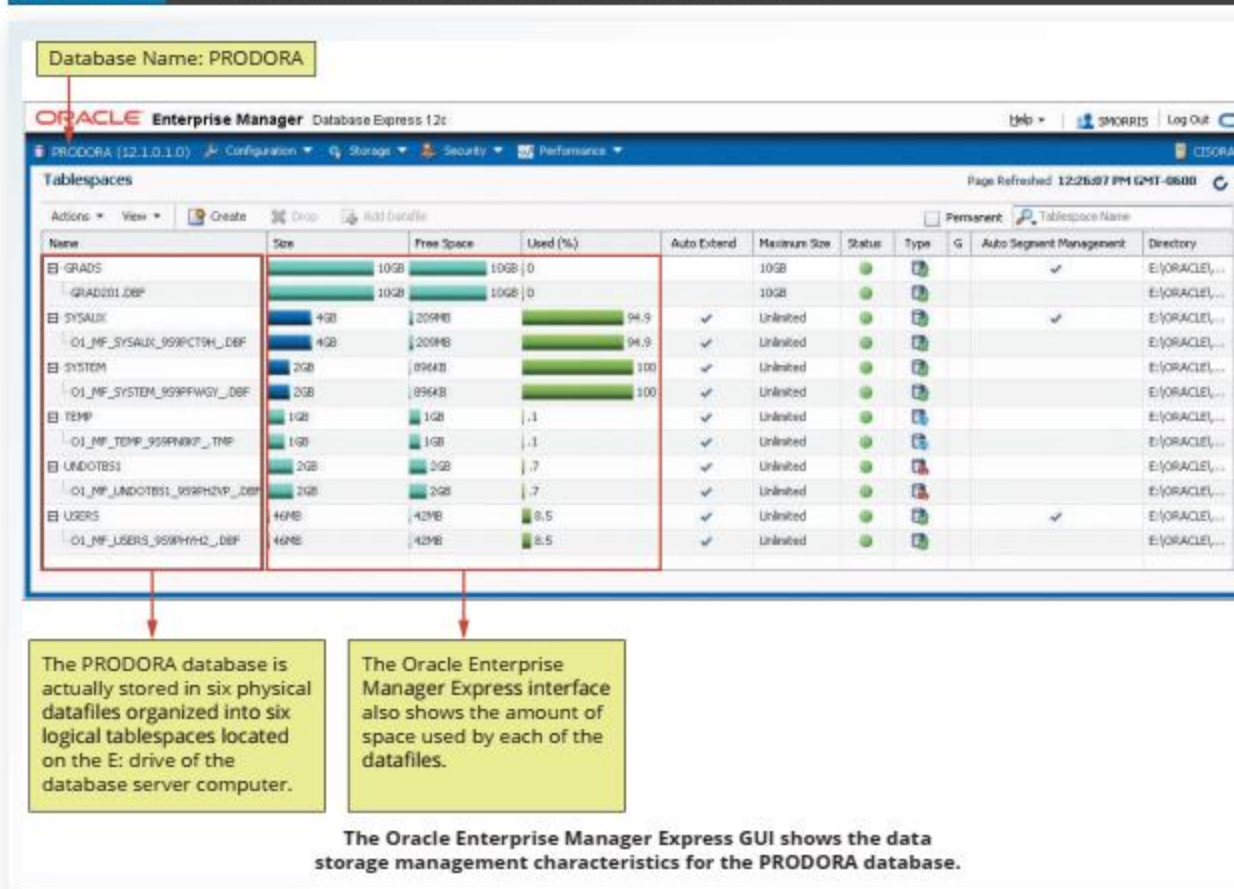


Figure 1.12 Illustrating Data Storage Management with Oracle



query language

A nonprocedural language that is used by a DBMS to manipulate its data. An example of a query language is SQL.

Structured Query Language (SQL)

A powerful and flexible relational database language composed of commands that enable users to create database and table structures, perform various types of data manipulation and data administration, and query the database to extract useful information.

application programming interface (API)

Software through which applications interact with each other transmitting data, messages, status, etc.

1-7b DBMS Functions (cont'd)

□ Data storage management

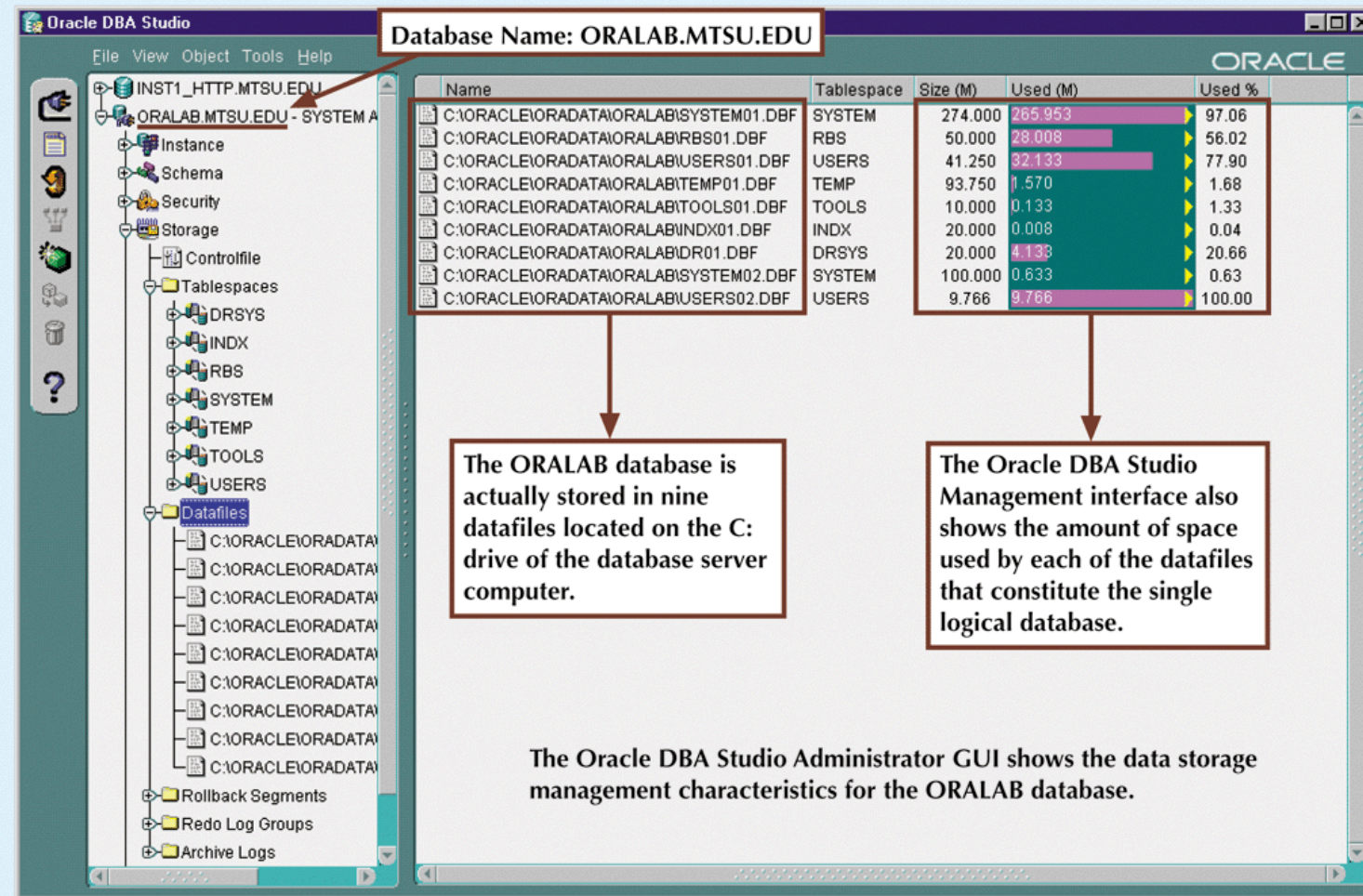
- DBMS creates and manages complex structures required for data storage
- Also stores related data entry forms, screen definitions, report definitions, etc.
- Performance tuning: activities that make the database perform more efficiently
- DBMS stores the database in multiple physical data files

Note

Why a Spreadsheet Is Not a Database

While a spreadsheet allows for the manipulation of data in a tabular format, it does not support even the most basic database functionality such as support for self-documentation through metadata, enforcement of data types or domains to ensure consistency of data within a column, defined relationships among tables, or constraints to ensure consistency of data across related tables. Most users lack the necessary training to recognize the limitations of spreadsheets for these types of tasks.

FIGURE 1.11 Illustrating data storage management with Oracle



SOURCE: Course Technology/Cengage Learning

DBMS Functions (cont'd.)

❑ Data transformation and presentation

- DBMS transforms data entered to conform to required data structures
- DBMS transforms physically retrieved data to conform to user's logical expectations

❑ Security management

- DBMS creates a security system that enforces user security and data privacy
- Security rules determine which users can access the database, which items can be accessed, etc.

DBMS Functions (cont'd.)

❑ Multiuser access control

- DBMS uses sophisticated algorithms to ensure concurrent access does not affect integrity

❑ Backup and recovery management

- DBMS provides backup and data recovery to ensure data safety and integrity
- Recovery management deals with recovery of database after a failure
 - ✓ Critical to preserving database's integrity

DBMS Functions (cont'd.)

□ Data integrity management

- DBMS promotes and enforces integrity rules
 - ✓ Minimizes redundancy
 - ✓ Maximizes consistency
- Data relationships stored in data dictionary used to enforce data integrity
- Integrity is especially important in transaction-oriented database systems

DBMS Functions (cont'd.)

- ❑ Database access languages and application programming interfaces
 - DBMS provides access through a query language
 - Query language is a nonprocedural language
 - Structured Query Language (SQL) is the de facto query language
 - ✓ Standard supported by majority of DBMS vendors

DBMS Functions (cont'd.)

□ Database communication interfaces

- Current DBMSs accept end-user requests via multiple different network environments
- Communications accomplished in several ways:
 - ✓ End users generate answers to queries by filling in screen forms through Web browser
 - ✓ DBMS automatically publishes predefined reports on a Web site
 - ✓ DBMS connects to third-party systems to distribute information via e-mail

1-7c Managing the Database System: A Shift in Focus

- ❑ Database system provides a framework in which strict procedures and standards enforced
 - Role of human changes from programming to managing organization's resources
- ❑ Database system enables more sophisticated use of the data
- ❑ Data structures created within the database and their relationships determine effectiveness

1-7c Managing the Database System: A Shift in Focus (cont'd.)

❑ Disadvantages of database systems:

- Increased costs
- Management complexity
- Maintaining currency
- Vendor dependence
- Frequent upgrade/replacement cycles

Preparing for Your Database Professional Career

Table 1.3 Database Career Opportunities

Job title	Description	Sample Skills Required
Database Developer	Create and maintain database-based applications	Programming, database fundamentals, SQL
Database Designer	Design and maintain databases	Systems design, database design, SQL
Database Administrator	Manage and maintain DBMS and databases	Database fundamentals, SQL, vendor courses
Database Analyst	Develop databases for decision support reporting	SQL, query optimization, data warehouses
Database Architect	Design and implementation of database environments (conceptual, logical, and physical)	DBMS fundamentals, data modeling, SQL, hardware knowledge, etc.
Database Consultant	Help companies leverage database technologies to improve business processes and achieve specific goals	Database fundamentals, data modeling, database design, SQL, DBMS, hardware, vendor-specific technologies, etc.
Database Security Officer	Implement security policies for data administration	DBMS fundamentals, database administration, SQL, data security technologies, etc.
Cloud Computing Data Architect	Design and implement the infrastructure for next-generation cloud database systems	Internet technologies, cloud storage technologies, data security, performance tuning, large databases, etc.
Data Scientist	Analyze large amounts of varied data to generate insights, relationships, and predictable behaviors	Data analysis, statistics, advanced mathematics, SQL, programming, data mining, machine learning, data visualization

Summary

- ❑ Data are raw facts
- ❑ Information is the result of processing data to reveal its meaning
- ❑ Accurate, relevant, and timely information is the key to good decision making
- ❑ Data are usually stored in a database
- ❑ DBMS implements a database and manages its contents

Summary (cont'd.)

- ❑ Metadata is data about data
- ❑ Database design defines the database structure
 - Well-designed database facilitates data management and generates valuable information
 - Poorly designed database leads to bad decision making and organizational failure
- ❑ Databases evolved from manual and computerized file systems

Summary (cont'd.)

- ❑ In a file system, data stored in independent files
 - Each requires its own management program
- ❑ Some limitations of file system data management:
 - Requires extensive programming
 - System administration is complex and difficult
 - Changing existing structures is difficult
 - Security features are likely inadequate
 - Independent files tend to contain redundant data
 - ✓ Structural and data dependency problems

Summary (cont'd.)

- ❑ Database management systems were developed to address file system's inherent weaknesses
- ❑ DBMS present database to end user as single repository
 - Promotes data sharing
 - Eliminates islands of information
- ❑ DBMS enforces data integrity, eliminates redundancy, and promotes security

- Data consists of raw facts. Information is the result of processing data to reveal its meaning. Accurate, relevant, and timely information is the key to good decision making, and good decision making is the key to organizational survival in a global environment.
- Data is usually stored in a database. To implement a database and to manage its contents, you need a database management system (DBMS). The DBMS serves as the intermediary between the user and the database. The database contains the data you have collected and “data about data,” known as metadata.
- Database design defines the database structure. A well-designed database facilitates data management and generates accurate and valuable information. A poorly designed database can lead to poor decision making, and poor decision making can lead to the failure of an organization.
- Databases can be classified according to the number of users supported, where the data is located, the type of data stored, the intended data usage, and the degree to which the data is structured.
- Databases evolved from manual and then computerized file systems. In a file system, data is stored in independent files, each requiring its own data management programs. Although this method of data management is largely outmoded, understanding its characteristics makes database design easier to comprehend.
- Some limitations of file system data management are that it requires extensive programming, system administration can be complex and difficult, making changes to existing structures is difficult, and security features are likely to be inadequate. Also, independent files tend to contain redundant data, leading to problems of structural and data dependence.
- DBMSs were developed to address the file system’s inherent weaknesses. Rather than depositing data in independent files, a DBMS presents the database to the end user as a single data repository. This arrangement promotes data sharing, thus eliminating the potential problem of islands of information. In addition, the DBMS enforces data integrity, eliminates redundancy, and promotes data security.
- Knowledge of database technologies leads to many career opportunities in the ever-expanding IT industry. There is a variety of specialization within the database arena for a wide range of skills and expertise.

Key Terms

ad hoc query	centralized database	data dictionary
analytical database	cloud database	data inconsistency
application programming interface (API)	data	data independence
business intelligence	data anomaly	data integrity
data processing (DP) specialist	data dependence	data management query
data quality	file	query language
data redundancy	general-purpose database	query result set
data type	information	record
data warehouse	islands of information	semistructured data
database	knowledge	single-user database
database design	logical data format	social media
database management system (DBMS)	metadata	structural dependence
database system	multiuser database	structural independence
desktop database	NoSQL	structured data
discipline-specific database	online analytical processing (OLAP)	Structured Query Language (SQL)
distributed database	online transaction processing (OLTP) database	transactional database
enterprise database	operational database	unstructured data
Extensible Markup Language (XML)	performance tuning	workgroup database
field	physical data format	XML database
	production database	

1. Define each of the following terms:
 - a. data
 - b. field
 - c. record
 - d. file
2. What is data redundancy, and which characteristics of the file system can lead to it?
3. What is data independence, and why is it lacking in file systems?
4. What is a DBMS, and what are its functions?
5. What is structural independence, and why is it important?
6. Explain the differences among data, information, and a database.
7. What is the role of a DBMS, and what are its advantages? What are its disadvantages?
8. List and describe the different types of databases.
9. What are the main components of a database system?
10. What is metadata?
11. Explain why database design is important.
12. What are the potential costs of implementing a database system?
13. Use examples to compare and contrast unstructured and structured data. Which type is more prevalent in a typical business environment?
14. What are some basic database functions that a spreadsheet cannot perform?
15. What common problems do a collection of spreadsheets created by end users share with the typical file system?
16. Explain the significance of the loss of direct, hands-on access to business data that end users experienced with the advent of computerized data repositories.
17. Explain why the cost of ownership may be lower with a cloud database than with a traditional, company database.